DANEBEN! SMART IOT WASTE BIN SYSTEM — PROTOTYPE BILL OF MATERIALS (BOM)

Document ID: BOM-IOT-2026-PROTO | Status: PRELIMINARY (BOM NOT FINAL) | Scope: Core Active Electronics & Carrier Board v2.0 Subsystems | Date: October 2026

Pos.	Component & Technical Specification	Website / Vendor	Order Code / SKU / MPN	Qty	Est. Unit Price	Material Specification / Notes
01	Seeed Studio XIAO ESP32-S3 Sense Dual-Core 240MHz, 8MB PSRAM, 8MB Flash, OV2640	**Farnell (de.farnell.com)** **(Alt: Reichelt.de)**	113991115 (Reichelt: XIAO ESP32S3 SEN)	**1×**	**€14.35** **(Alt: €15.00)**	Sense daughterboard with OV2640 camera (MPN: 113991115)
02	Waterproof Ultrasonic Sensor Module (JSN-SR04T) IP67 sealed probe, 40kHz, 2.5m coaxial cable	**Funduino.com** **(Alt: Roboter-Bausatz.de)**	AJ-SR04M / JSN-SR04T (RBS: RBS11632)	**2×**	**€6.21** **(€12.42 for 2)**	2.5m sealed coaxial cable with 2-pin connector
03	MAX98357A I2S Class-D Audio Amplifier Mono DAC/Amp breakout, 3.2W @ 4Ω, 2.54mm pitch	**Funduino.com** **(Alt: BerryBase.de)**	MAX98357 Audioverstärker (BerryBase: ADA5770)	**1×**	**€4.08** **(Alt: €4.90)**	Single-channel Class-D breakout, 2.54mm pitch
04	Miniature Dynamic Speaker (Ø 40 mm) Ø 40mm, 3W, 4Ω / 8Ω, pre-soldered leads	**Funduino.com** **(Alt: BerryBase.de / Reichelt)**	Lautsprecher 3W 4 Ohm (Reichelt: Ekulit AL-23P)	**1×**	**€2.40** **(Alt: €2.50)**	—
05	Daneben Prototype Carrier Board PCB v2.0 Custom 2-layer FR4 motherboard (130 × 90 mm)	**JLCPCB / PCBWay** **(Alt: In-House Fabrication)**	DANEBEN-CARRIER-V2 (Gerbers in hardware/)	**1×**	**€3.00** **(Batch of 5: €15.00)**	FR4 1.6mm, 2-layer, 1 oz Cu, lead-free HASL
06	Tactile Calibration Pushbuttons (12 × 12 mm) 12×12×7.3mm, 4-pin THT, 50mA / 12V DC	**BerryBase.de** **(Alt: Reichelt.de)**	Kurzhubtaster 12x12mm (Reichelt: TASTER 1212)	**2×**	**€0.35** **(€0.70 for 2)**	—
07	Status LEDs & Current-Limiting Resistors 3× THT LEDs (Amber, Blue, Green) + 3× 1kΩ	**BerryBase.de** **(Alt: Reichelt.de)**	LED 3mm/5mm Set + 1k Ohm (Reichelt: LED 3MM RT/GN/GE)	**1 Set** **(3× LEDs)**	**€0.60** **(Kit)**	—
08	Master Power Slide Switch (SPDT) 3-pin through-hole slide switch, 2.54mm pitch	**BerryBase.de** **(Alt: Reichelt.de)**	Schiebeschalter Print 1-polig (Reichelt: T 215)	**1×**	**€0.45**	—
09	USB-C Power Receptacle + CC Resistors & Filter Cap Type-C 16-Pin female + 2× 5.1kΩ + 100µF 16V Elko	**BerryBase.de** **(Alt: Amazon.de Prime)**	USB-C 16-Pin + 100µF Elko (Amazon: Type-C 16P)	**1 Set**	**€0.90** **(Set)**	—
10	PCB Screw Terminal Blocks (5.08 mm Pitch) 2-pin wire-to-board modular blocks, 10A / 300V	**BerryBase.de** **(Alt: Reichelt.de)**	Schraubklemme 2-polig RM 5.08 (Reichelt: AKL 055-02)	**2×**	**€0.40** **(€0.80 for 2)**	—
11	Female Socket Header Kit (2.54 mm Pitch) Machine-tooled sockets (MCU, Amp, Sensors)	**BerryBase.de** **(Alt: Reichelt.de)**	Buchsenleisten 2.54mm (Reichelt: BL 1X36G)	**1 Set**	**€1.05** **(Kit)**	—
Σ TOTAL PROTOTYPE HARDWARE (GROSS INCL. 19% VAT)				**16×**	**€40.75** **(Pure Parts Total)**	Complete prototype BOM (Core + Carrier v2)

** NOTE: BoM not final — Preliminary prototype specification. Material specifications, vendor part numbers, and pricing subject to laboratory validation.*